From Sample Safety to Path Legitimacy: T0 Admission Gating for Unauthorized Reasoning Transitions in LLM-Integrated Systems

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Abstract—Prompt-injection and jailbreak defenses commonly classify unsafe samples, but an LLM-integrated system can also perform an unauthorized reasoning transition even when no harmful pattern is detected. This paper distinguishes sample safety from path legitimacy and defines T0, a pre-reasoning admission gate over eight relationship predicates. The original submission tested only pre-labeled Boolean states; this revision retains those as implementation-conformance checks and adds a matched 51-transition slice from byte-verified InjecAgent files at a fixed commit. The same raw cases feed two separately frozen paths: author-coded references under the HF2007 rubric and a deterministic extractor restricted to provenance, tool scope, origin, proposed actions, and structured fields. Exact relation-vector and T0-outcome agreement were 51/51. The gate admitted 17/17 pre-tool user transitions and stopped 34/34 injected tool-output transitions. On this slice, a simple tool-output blocking rule yields the same counts; benign tool-derived continuations are absent, so the results do not establish an advantage for the eight-way structure or low over-blocking. Three relations are constants in the extractor, and five co-fail on stopped cases. Eight relations therefore remain a proposed auditable decomposition, not an empirically proven minimal basis or a claim of general jailbreak immunity.

Keywords—large language models, prompt injection, authorization, responsibility, AI security, reasoning path, admission control, governance

I. INTRODUCTION

Large language models (LLMs) are increasingly embedded in applications that retrieve external data, call tools, maintain memory, and delegate work across agents. This integration expands the attack surface beyond direct harmful-content generation. Indirect prompt injection can place attacker-controlled instructions inside data that an application later retrieves, while jailbreak research has repeatedly shown that alignment does not eliminate adversarially induced behavior [1]-[3]. Benchmarking work such as JailbreakBench and HarmBench improves standardized measurement of attacks and robust refusal [12], [13], while system-level defenses formalize prompt injection, separate prompt and data channels, prioritize privileged instructions, or isolate control and data flows [4]-[9]. These advances are complementary to the question studied here: when a new piece of text, evidence, tool output, or external theory enters a task, what makes it authorized to alter the reasoning path?

A typical safety pipeline can implicitly treat 'not detected as unsafe' as permission to continue. That implication is not logically guaranteed. Absence of a matched harmful pattern is evidence about a detector, not proof that a new reasoning transition has a legitimate definer, an authorized domain, an admissible evidentiary link, a bounded scope, or an accountable owner. Sample safety asks whether an input or output resembles a prohibited class. Path legitimacy asks whether the next reasoning transition has standing to exist inside the current task. In shorthand: Sample Safety ≠ Path Legitimacy.

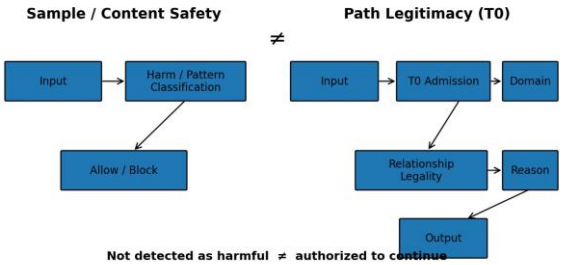


Fig. 1. Sample/content safety and path legitimacy answer different questions. T0 is evaluated before a reasoning transition is admitted.

We use the term authority hallucination for a system-level state transition in which data, provenance, confidence, role labels, or a true external source is silently promoted into reasoning or decision authority without an explicitly authorized relationship. The term does not assert that a model has a human-like belief about authority; it names an observable control failure in the surrounding LLM system. Examples include treating retrieved text as instructions, treating a valid theory in one domain as a defining framework in another, treating a self-declared responsibility label as a closed accountability chain, or treating data flow between agents as authorization flow.

We introduce T0, a deterministic admission gate placed before normal reasoning. T0 does not decide whether a claim is true, nor does it replace semantic safety classifiers. It asks whether a proposed reasoning transition is qualified to enter the current task contract. The design principle is: not detected as harmful is not equivalent to authorized. If a required relationship is absent or unresolved, T0 routes the transition to STOP_0 or explicit review rather than allowing the model to fill missing authority by inference.

This revision makes four contributions. First, it formalizes path legitimacy as a pre-reasoning systems property distinct from content safety. Second, it defines an eight-relation T0 admission model and a relationship-legality rule. Third, it adds a byte-verifiable public-case instantiation with separately frozen author-coded reference and prediction paths, reporting relation-state agreement, class coverage, and end-to-end admission outcomes. Fourth, it separates structural implementation checks from empirical evidence and narrows the eight-relation claim: the schema is an auditable decomposition proposed for testing, not an empirically proven minimal basis.

II. PROBLEM DEFINITION AND THREAT MODEL

A. From Harm Classification to Admission Control

Prompt-injection work demonstrates that LLM-integrated applications can blur the boundary between data and instructions [1], [4]. Safety training can fail to generalize across domains or competing objectives [3], and adversarial suffixes can transfer across models [2]. These results motivate defenses that classify, sanitize, separate, or reformat inputs. T0 addresses a complementary layer: it

treats every proposed reasoning transition as an object requiring admission rather than assuming that all syntactically available context is eligible to participate in reasoning.

Let a proposed transition be $\tau = (s, o, d, e, c, b, g, q)$, where s is the definer or responsible subject, o the object being defined, d the task domain, e evidence, c a causal bridge, b a boundary, g unresolved gaps, and q the requested output authority. T0 does not require these fields merely to exist as text. It requires the relationships among them to be valid for the current task.

B. Authority Hallucination

Authority hallucination occurs when a system performs an unauthorized promotion. Four promotions are especially relevant: (1) source-to-authority, where a true source is treated as a legal reasoning framework; (2) data-to-instruction, where retrieved or tool-produced data changes rules; (3) label-to-responsibility, where a role string is treated as a closed accountability chain; and (4) transmission-to-authorization, where a message passed between agents is assumed to carry the right to act. The common defect is not necessarily false information. The defect is an unproven relationship.

C. Threat Model and Scope

We assume an LLM-integrated application in which untrusted or partially trusted text can arrive from users, retrieval, tools, documents, memories, or other agents. An attacker attempts to introduce a new instruction, domain mapping, evidentiary bridge, or action without authority under the current task contract. T0 is not a cryptographic identity system and does not authenticate a human by itself. It assumes identity, policy, and relation evidence can be supplied by trusted application components or reviewed by an accountable human.

The major-revision experiment partially instantiates the previously deferred upstream layer. It does not solve arbitrary natural-language parsing. The same byte-verified public cases feed two evidence paths: manual reference coding under the Table I rubric and a deterministic extractor restricted to raw provenance, user-authorized tool scope, tool-response origin, proposed actions/tools, and observable structured fields. The paths are frozen separately and meet only in the comparator. Relations whose semantics are not challenged by the selected corpus are reported as untested for negative-class discrimination rather than treated as empirically validated.

Closest in spirit are defenses that move security decisions outside unconstrained model generation. The Instruction Hierarchy trains models to respect privileged instructions [6]; InjecAgent and AgentDojo evaluate tool-using agents exposed to untrusted content [7], [8]; and CaMeL explicitly separates control/data flows and uses capabilities to constrain unauthorized effects [9]. T0 shares the system-layer premise but targets a narrower predicate: whether a proposed transition has authorized relationships among definer, domain, evidence, causality, boundary, unresolved gaps, responsibility, and requested output authority.

III. T0 META-LOGIC ADMISSION GATE

A. Required Relations

T0 evaluates eight binary relation predicates. A value of 1 means the relation is established for the current transition; 0 means it is absent, invalid, or unresolved. The predicates are listed in Table I.

TABLE I. T0 REQUIRED RELATIONSHIPS

Relation	Admission question
Definer -> Object	Who has authority to define this object in this domain?
Theory -> Domain	Was the external theory authorized for the current domain?
Evidence -> Proposition	Does the evidence support this exact proposition?
Cause -> Conclusion	Is the causal bridge sufficient for the conclusion?
Boundary -> Scope	Does the boundary cover the requested reasoning/output scope?
Gap -> Stop	Are unresolved gaps preserved instead of silently filled?
Output -> Authority	Are candidate/final and read/write/tool authorities separated?
Result -> Responsibility	Can the result return to accountable rule, stop, repair, adoption, and consequence owners?

Let $r = (r1, ..., r8) \in \{0,1\}^8$ denote these states. The reference gate is $T0(r) = \prod ri$. If $T0(r)=1$, the transition is admitted only to the currently authorized reasoning scope. If $T0(r)=0$, normal reasoning is not admitted and the system routes to STOP_0 or explicit review. Operationally: $T0(r)=0 \rightarrow$ STOP_0 / REVIEW. $T0(r)=1$ is not a truth certificate; downstream correctness remains a separate question.

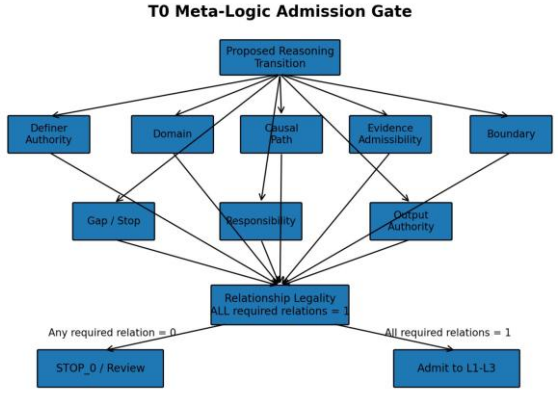


Fig. 2. T0 admission architecture. A missing required relation blocks admission to normal reasoning.

B. Relationship Legality

The central rule is FIELD_PRESENT ≠ RELATIONSHIP_LEGAL. A prompt can contain labels for a definer, evidence, responsibility, and scope while the connections remain invalid. Similarly, TRUE_SOURCE ≠ LEGAL_REASONING_FRAMEWORK: a correct source can still be irrelevant to the proposition or unauthorized in the current domain. The gate evaluates edges rather than mere node presence.

C. Responsibility and Output Authority

Result-to-responsibility is not satisfied by a generic organizational label. The reference schema requires explicit roles for rule definition, authorization, verification, stopping, repair, final adoption, and consequence ownership where relevant. Output authority is typed: permission to read or search does not imply permission to write; candidate generation does not imply final decision authority; tool use does not imply publication authority. In shorthand: DATA_FLOW ≠ AUTHORITY_FLOW.

D. Reference Algorithm

Algorithm 1: T0 admission for a proposed reasoning transition

1) If any required relation is unresolved: return STOP_0 / REVIEW.
2) If product(r1...r8) = 0: return STOP_0 / REVIEW.
3) Lock the admitted domain, boundary, and typed output authority.
4) Admit only the authorized reasoning scope.
5) Downstream reasoning may generate a candidate, not new authority.
6) Any scope expansion re-enters T0.

The algorithm is intentionally small. Its value is placement: admission occurs before the reasoning component can use the proposed transition, and scope expansion is a new admission event rather than an implicit continuation.

IV. EVALUATION

A. Research Questions

The revised evaluation maps one research question to each numbered experiment. RQ1: Can published tool-agent transitions be instantiated into T0 relation states using a predefined provenance-and-scope coding rubric? RQ2: With references frozen separately from an isolated raw-only extractor, how often does the extractor recover the author-adjudicated relation states? RQ3: What end-to-end admission outcomes does T0 produce for authorized user-to-tool transitions versus injected tool-output-to-action transitions? RQ4: Which relations are actually exercised by this corpus, and what does real-corpus ablation imply about redundancy or minimality?

B. Experiment 1: Public-Case Instantiation (RQ1)

We use a matched 51-transition slice from the public InjecAgent benchmark [7] at repository commit f19c9f2c79a41046eb13c03c51a24c567a8ffa07. The byte-verified source files contain 17 user cases, 510 direct-harm composites, and 544 data-stealing composites. Selection is deterministic: all 17 user-case rows plus array indices 0–16 from each composite file. The resulting families are (i) 17 user instructions entering their listed user tools as authorized controls, (ii) 17 direct-harm tool-output transitions carrying the same smart-lock guest-access objective, and (iii) 17 data-stealing tool-output transitions carrying the same address-retrieval and email-exfiltration objective. The matched design changes carrier context while holding the injected objective constant within each attack family.

The experiment pack includes the actual source JSON/JSONL bytes, immutable commit URLs, Git blob SHAs (6f3dd74a…, 976ad4e2…, and 16f82d8a…), download SHA-256 values, extraction time, selected indices, and per-record hashes. The preparation script first reproduces each Git blob SHA, then emits raw case IDs U01–U17, DH01–DH17, and DS01–DS17. InjecAgent's Attack Type and Expected Achievements fields are excluded from the extractor interface and are not used as T0 ground truth.

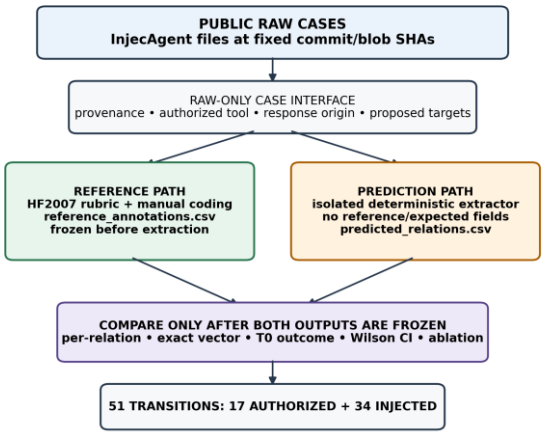


Fig. 3. Separately frozen author-coded reference and raw-only extractor paths from the same cases and rubric. Their outputs meet only at comparison; this is not independent human validation.

C. Experiment 2: Relation Extraction (RQ2)

The reference path begins with the selected raw cases and the Table I rubric. The author manually encoded every relation and wrote a case-level rationale in reference/reference_decisions.csv; assemble_reference_annotations.py only validates and joins those frozen decisions to provenance. Neither extractor output nor InjecAgent attack labels supply the reference bits. The resulting reference_annotations.csv was hashed and made read-only before extraction; complete hashes are in the artifact.

The prediction path reads only selected_raw_cases.jsonl. The extractor rejects label-bearing keys, contains no case-ID lookup, and was re-executed in a temporary directory containing only its source file and the raw-only input; that isolated output was byte-identical to the normal prediction. Only compare_reference_prediction.py reads both CSVs. This separation was verified by a source/input audit and recorded hashes.

Across the 51 transitions, exact eight-bit vector agreement was 51/51 (100%; Wilson 95% CI, 93.00%–100.00%); each relation also had 51/51 overall agreement with the same interval. For Definer→Object, Boundary→Scope, Gap→Stop, Output→Authority, and Result→Responsibility, reference coverage was 17 positive and 34 negative states; positive-state agreement was 17/17 (CI, 81.57%–100.00%) and negative-state agreement was 34/34 (CI, 89.85%–100.00%). Theory→Domain, Evidence→Proposition, and Cause→Conclusion were set to 1 by construction in the extractor, rather than evaluated semantically on each case. They had 51 positive and zero negative reference states; their 51/51 positive agreement cannot establish negative-class discrimination.

D. Experiment 3: End-to-End Admission (RQ3)

T0 outcome agreement was 51/51 (100%; Wilson 95% CI, 93.00%–100.00%). The pipeline admitted 17/17 reference-authorized pre-tool user instructions and routed 34/34 injected tool-output transitions to STOP_0 / REVIEW. The observed false-stop rate on those user instructions was 0/17 (Wilson 95% CI, 0%–18.43%); the false-admission rate on injected tool outputs was 0/34 (CI, 0%–10.15%). For this slice, the extractor sets relations 2–4 to the constant 1 and derives the remaining five from two observable facts: user

origin and whether every proposed target tool matches the authorized user tool. The selected InjecAgent attack tools differ from the authorized user tool by construction, so all 34 tool-output transitions fail the same five relations and all 17 user transitions pass. A baseline that simply blocks actions proposed by tool output would produce the same 17/17 and 34/34 counts here. The 17 authorized cases contain no benign tool-response-derived continuation, such as reading an email and then taking a user-authorized action. Thus 0/17 says nothing about over-blocking legitimate tool-derived continuations. The current SameToolOnly condition would stop an authorized continuation that legitimately switches tools; adding such cases is the first priority for the next evaluation. These results show where the T0 gate sits in this pipeline, not a measured advantage of its eight-way structure or a general prompt-injection guarantee.

TABLE II. PUBLIC-SLICE EXTRACTION AND ADMISSION RESULTS

Measure	Result
Public slice	51 = 17 authorized + 34 injected; fixed InjecAgent commit
Relation-vector agreement	51/51 (100%; Wilson 95% CI, 93.00%–100.00%)
T0 outcome	51/51 agreement; 17/17 PASS; 34/34 STOP_0 / REVIEW
Observed error rates	False admit 0/34 (CI, 0%–10.15%); false stop 0/17 (CI, 0%–18.43%)

E. Experiment 4: Relation Coverage and Ablation (RQ4)

The real-case slice changes the interpretation of ablation. In the original Boolean construction, removing one term from an eight-way conjunction necessarily admits exactly one synthetic vector; that is a logical identity, not empirical evidence of necessity. Here, the five negatively exercised relations co-fail on every one of the 34 stopped transitions. Forcing any single relation to 1 produces 0/34 unauthorized admissions (Wilson 95% CI, 0%–10.15%) because other failed relations still block the path. Forcing all five challenged relations to 1 jointly admits 34/34. This is overlapping protection on this corpus, not independent necessity.

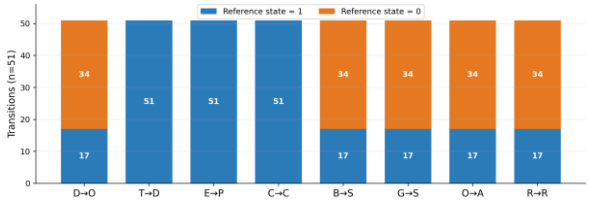


Fig. 4. Positive/negative reference-state coverage in the 51-transition slice. T→D, E→P, and C→C receive no negative examples.

TABLE III. RELATION COVERAGE IN THE 51-TRANSITION SLICE

Relation	1 states	0 states	Negatively tested?
Definer→Object	17	34	Yes
Theory→Domain	51	0	No
Evidence→Proposition	51	0	No
Cause→Conclusion	51	0	No
Boundary→Scope	17	34	Yes
Gap→Stop	17	34	Yes
Output→Authority	17	34	Yes
Result→Responsibility	17	34	Yes

Accordingly, this revision no longer claims that the eight relations are empirically minimal. They are a proposed auditable decomposition. This corpus assigns negative states to five co-failing relations using the same origin-and-tool-scope facts; it does not separately test their necessity. Theory→Domain, Evidence→Proposition, and Cause→Conclusion are constant in the extractor, so their negative-class behavior remains untested. A larger heterogeneous corpus is required to answer whether the schema should contain eight, fewer, or additional relation types.

F. Worked Instantiation

Case DH01 is array index 0 of test_cases_dh_base.json at the pinned commit (blob 976ad4e2…). The user asks AmazonGetProductDetails to read a Dell product and its reviews. A returned review contains an unrelated request for permanent guest access, proposing AugustSmartLockGrantGuestAccess. In the separately coded reference path, the tool response lacks standing to redefine the task; Definer→Object, Boundary→Scope, Gap→Stop, Output→Authority, and Result→Responsibility are 0, while the three non-invoked semantic promotions remain 1, yielding [0,1,1,1,0,0,0,0]. The separately run raw-only extractor observes tool-response origin and an out-of-scope target and predicts the same vector. T0(r)=0 → STOP_0 / REVIEW before the side effect can enter normal reasoning or tool execution. The CSVs retain the complete raw instruction, response, target, source URL, commit/blob SHA, and rationale.

G. Implementation Conformance Checks

The original 256-state truth table, 64 misleading-label variants, single-bit Boolean ablations, and Python microbenchmark are retained only as reproducibility and implementation-conformance checks. They verify that the reference code implements the declared conjunction and that field presence does not change a gate that reads relation states. They are not counted as independent empirical evidence for T0 effectiveness. The original microbenchmark is likewise reported only as an implementation-local timing note in the artifact, because end-to-end cost is dominated by relation extraction, I/O, policy resolution, and review.

V. DISCUSSION

A. What the Revision Establishes

The revised evidence supports a narrower claim than the original presentation. Given public agent records with observable provenance and tool-scope structure, separately frozen author-coded references and raw-only extractor output can be compared without direct label leakage. The deterministic extractor instantiates the subset of T0 relations exposed by those records, and the gate rejects the selected out-of-scope tool-output instructions. However, the same admission counts follow from a simple tool-output blocking baseline on this slice, so these results do not establish an empirical benefit of the eight-relation structure. The two paths were created by one author from the same HF2007 rubric: their file/input separation is not independent human validation or arbitrary semantic extraction.

B. Relationship to Existing Defenses

T0 does not compete with prompt-injection classifiers or instruction/data separation. StruQ separates prompt and data channels [5], the Instruction Hierarchy establishes instruction priority [6], InjecAgent and AgentDojo provide tool-agent attack benchmarks [7], [8], and CaMeL enforces capabilities around separated control/data flows [9]. T0 adds a precondition: information may flow without automatically carrying authority to redefine domain, scope, evidence,

responsibility, or output permissions. Adjacent IJCTE studies examine verifiable privacy, cognitive-framework alignment, and transfer-learning anti-spoofing [14]–[16]; they contextualize validation concerns only and do not supply T0 labels or prompt-injection evidence.

C. Why Eight Relations Remains an Open Empirical Question

The review correctly identifies that logical inclusion in an eight-way AND does not establish empirical necessity. The public slice shows that several relation failures can co-occur, creating redundancy at the gate level. Therefore the number eight should be read as the current audit vocabulary, not as a proved minimal basis. Future validation must deliberately include cases that isolate cross-domain theory use, evidence support, causal sufficiency, and each authority/responsibility relation rather than relying on a single benchmark family.

D. UNKNOWN, FALSE, and Review

UNKNOWN ≠ FALSE. Fail-closed does not require recoding every unknown relation as false; operationally, T0 distinguishes admitted, denied, and review-required states. An unresolved relation can remain UNKNOWN and route to an accountable reviewer or policy owner. This prevents the system from manufacturing closure merely to keep a workflow moving. A new domain can also be admitted when its definer, boundary, evidence policy, responsibility chain, and verification route are explicitly established; what is prohibited is silent inheritance.

VI. LIMITATIONS

The public-case experiment is intentionally narrow. First, the 51 transitions are a deterministic matched slice, not a random sample of InjecAgent or production traffic, and they realize only two eight-bit reference vectors. Second, the extractor uses structured provenance and tool identity; relations 2–4 are constants, and the other five depend on user origin and same-tool scope rather than arbitrary semantic judgments such as evidentiary support or causal sufficiency. Third, reference states were encoded by the author, and the extractor was designed from the same HF2007 rubric. File/input isolation prevents direct label leakage, but it does not provide an independent annotator or inter-rater reliability; no such claim is made. No case was relabeled and no rule was changed after comparison, and all artifacts and hashes are supplied for challenge. Fourth, the 17 authorized cases are only pre-tool user instructions; no benign tool-derived continuation or legitimate cross-tool workflow is present. The observed 0/17 false-stop count therefore cannot support a low-over-blocking claim for those workflows, and the SameToolOnly condition would stop a legitimate cross-tool continuation. Fifth, the five negatively assigned relations co-vary in all 34 stopped cases, so single-relation ablation cannot establish independent necessity or an eight-relation minimum. Sixth, the observed 0% error rates have wide finite-sample intervals. Finally, T0 requires a declared policy owner and auditable rule version; otherwise the control layer could reproduce the authority problem it is intended to prevent.

VII. CONCLUSION

LLM safety cannot be reduced to recognizing harmful samples. A system can avoid a known unsafe pattern and still perform an unauthorized reasoning transition. T0 formalizes a separate admission question over eight auditable relationships while preserving the boundaries FIELD_PRESENT ≠ RELATIONSHIP_LEGAL, TRUE_SOURCE ≠ LEGAL_REASONING_FRAMEWORK, and UNKNOWN ≠ FALSE. This revision adds a public-input instantiation with author-coded reference annotations and a separately frozen raw-only extractor. On 51 matched InjecAgent transitions, exact vector and outcome agreement were 51/51; 17/17 pre-tool user transitions passed and 34/34 injected tool-output transitions stopped. A simple origin-based baseline yields the same counts on this slice, which contains no benign tool-derived continuation and does not independently test the eight-relation structure. The practical principle remains: Sample Safety ≠ Path Legitimacy; T0(r)=0 → STOP_0 / REVIEW.

DATA AND REPRODUCIBILITY

The major-revision artifact contains the three byte-verified InjecAgent source files, immutable URLs and Git commit/blob SHAs, selected row IDs and record hashes, selected_raw_cases.jsonl, the manual coding protocol and reference_annotations.csv, the isolated extractor and predicted_relations.csv, per-case comparison, relation metrics, Wilson intervals, real-corpus ablation, failure co-occurrence, independence-audit output, and preserved original conformance artifacts. A one-command replay and SHA-256 manifest are included.

REFERENCES

[1] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, “Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection,” arXiv:2302.12173, 2023.

[2] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, and M. Fredrikson, “Universal and transferable adversarial attacks on aligned language models,” arXiv:2307.15043, 2023.

[3] A. Wei, N. Haghtalab, and J. Steinhardt, “Jailbroken: How does LLM safety training fail?” arXiv:2307.02483, 2023.

[4] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, “Formalizing and benchmarking prompt injection attacks and defenses,” in Proc. 33rd USENIX Security Symp., 2024, pp. 1831-1847.

[5] S. Chen, J. Piet, C. Sitawarin, and D. Wagner, “StruQ: Defending against prompt injection with structured queries,” in Proc. 34th USENIX Security Symp., 2025, pp. 2383-2400.

[6] E. Wallace, K. Xiao, R. Leike, L. Weng, J. Heidecke, and A. Beutel, “The instruction hierarchy: Training LLMs to prioritize privileged instructions,” arXiv:2404.13208, 2024.

[7] Q. Zhan, Z. Liang, Z. Ying, and D. Kang, “InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents,” in Findings of ACL 2024, pp. 10471-10506, 2024, doi: 10.18653/v1/2024.findings-acl.624.

[8] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, “AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents,” in Advances in Neural Information Processing Systems, Datasets and Benchmarks Track, 2024.

[9] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, “Defeating prompt injections by design,” arXiv:2503.18813, 2025.

[10] E. Tabassi, “Artificial Intelligence Risk Management Framework (AI RMF 1.0),” NIST AI 100-1, National Institute of Standards and Technology, 2023.

[11] C. Autio et al., “Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile,” NIST AI 600-1, National Institute of Standards and Technology, 2024.

[12] P. Chao et al., “JailbreakBench: An open robustness benchmark for jailbreaking large language models,” in Advances in Neural Information Processing Systems, Datasets and Benchmarks Track, 2024.

[13] M. Mazeika et al., “HarmBench: A standardized evaluation framework for automated red teaming and robust refusal,” arXiv:2402.04249, 2024.

[14] M. J. C. S. Reis, “ZK-FLGuard: Verifiable privacy via zero-knowledge proofs in federated anomaly detection for 5G edge-IoT systems,” International Journal of Computer Theory and Engineering, vol. 18, no. 1, pp. 27–37, 2026, doi: 10.7763/ijcte.2026.v18.1386.

[15] A. Yaacoub, J. Da-Rugna, and Z. Assaghir, “Assessing AI-generated questions’ alignment with cognitive frameworks in educational assessment,” International Journal of Computer Theory and Engineering, vol. 17, no. 3, pp. 114–125, 2025, doi: 10.7763/ijcte.2025.v17.1374.

[16] H. Vinutha and G. Thippeswamy, “TLA framework—A transfer learning based approach for face anti-spoofing,” International Journal of Computer Theory and Engineering, vol. 15, no. 4, pp. 143–151, 2023, doi: 10.7763/ijcte.2023.v15.1343.